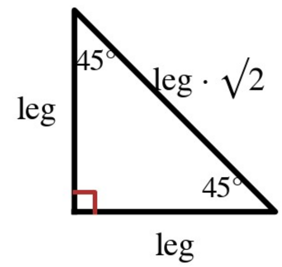
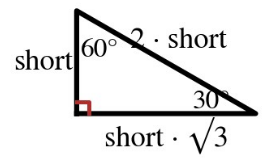
SC Honors Geometry - Nappier
9.2 Special Right Triangles

Speak **L**ike **A** **M**athematician:

Where do the two special right triangles come from?





Special Right Triangle Relationships

45°-45°-90°: the hypotenuse is ________ times as long as each leg (and the two legs are congruent).

30°-60°-90°: the hypotenuse is ________ times the SHORTER leg, and the longer leg is ________ times the shorter leg.

The shorter leg is always opposite the ________° angle; the longer leg is opposite the ________° angle.

Example 1 – Rationalize Each Denominator (Algebra 1 recall)

a) $5/\sqrt{3}$ and $7/\sqrt{2}$

b) $4/\sqrt{8}$ and $6/\sqrt{12}$

c) $\sqrt{3}/\sqrt{6}$

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Example 2 – 45°-45°-90°

Give every answer in simplest radical form. Figures at the right.

a) Legs 8. Find the hypotenuse x.

b) Hypotenuse 10√2. Find each leg x.

c) Hypotenuse 14. Find each leg x.

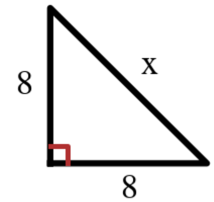
Example 3 – 30°-60°-90°

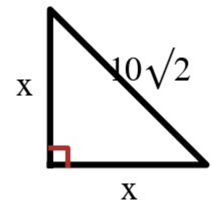
In each figure, the shorter leg is opposite the 30° angle.

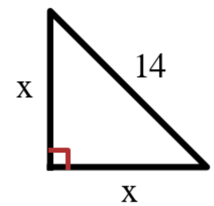
a) Shorter leg 5. Find the hypotenuse x and longer leg y.

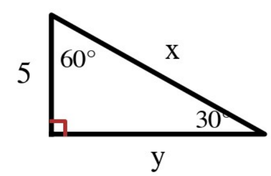
b) Longer leg 12. Find the shorter leg x and hypotenuse y.

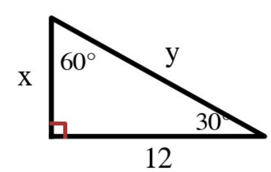
c) Hypotenuse 18. Find the shorter leg x and longer leg y.

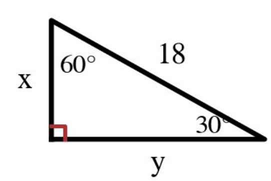






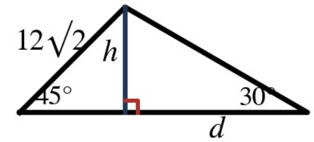






Example 4 – A Composite Figure

In the figure at the right, the altitude h splits the big triangle into a 45°-45°-90° triangle (left) and a 30°-60°-90° triangle (right). The left hypotenuse is 12√2.



a) Find h.

b) Find d.

Example 5 – Special Triangles Hiding in Shapes

a) A square has diagonal 20. Find its side length (exact).

b) An equilateral triangle has side 12. Find its altitude (exact).

c) Why do these two shapes ALWAYS hide special right triangles?